

ACTL2102 Tutorial

Week 2: Discrete-Time Markov Chains

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Plan

- ▶ Main questions: **2.1, 2.2, 2.4 (parts 1–5), 2.10, 2.12**
- ▶ If time permits: 2.9

Three More Simulation Methods

Each avoids inverting a c.d.f. by using a distributional relationship instead.

Normal, by Box–Muller. If U_1, U_2 are independent $U(0, 1)$, then

$$X = (-2 \ln U_1)^{1/2} \cos(2\pi U_2), \quad Y = (-2 \ln U_1)^{1/2} \sin(2\pi U_2)$$

are independent standard normals, and $\mu + \sigma X$ is $N(\mu, \sigma^2)$.

Gamma(n, λ). A sum of n independent $\text{Exp}(\lambda)$ variables, so

$$X = -\frac{1}{\lambda} \sum_{i=1}^n \ln(U_i).$$

Here λ is a rate, so the mean is n/λ . Some references use the scale $1/\lambda$ instead.

Chi-squared. $\chi_n^2 = \sum_{i=1}^n Z_i^2$ with $Z_i \sim N(0, 1)$ independent.

Stochastic Processes and the Markov Property

A **stochastic process** is a collection of random variables $\{X(t), t \in T\}$: T is the **index set**, usually time, the values lie in the **state space** S , and one realisation of the whole process is a **sample path**.

The Markov property

$$\Pr(X_{n+1} = j \mid X_n = i, \dots, X_1 = i_1) = \Pr(X_{n+1} = j \mid X_n = i)$$

for all states and all n : given the present, the past adds nothing.

Read n as **now**: X_n is where the chain is, X_{n+1} where it goes next, and $X_1, \dots, X_{n-1}$ the history you are free to forget.

It is usually written $\Pr(X_{n+1} \mid X_n, \dots, X_1) = \Pr(X_{n+1} \mid X_n)$, meaning the line above at every choice of states.

Discrete-Time Markov Chains

A **Markov chain** is a Markov process with a **discrete state space** S . Module 2 is the discrete-time case, a **DTMC**, where X_n is the state after n steps. Module 4 keeps the discrete state space and lets time run continuously, a **CTMC**: the state space is discrete in both, only time differs.

The **one-step transition probabilities** are

$$P_{ij}^{(n,n+1)} = \Pr(X_{n+1} = j \mid X_n = i),$$

and if they do not depend on n the chain is **homogeneous**, written P_{ij} .

The transition matrix

$\mathbf{P} = (P_{ij})$ collects them: $P_{ij} \geq 0$, and **every row sums to 1**. Rows are “from”, columns are “to”. Checking row sums catches most errors immediately.

States are numbered from 0 in the notes and from 1 in some exercises. Either is fine; keep one order throughout.

Chapman–Kolmogorov

The **n -step transition probability** $P_{ij}^{(n)} = \Pr(X_{n+m} = j \mid X_m = i)$ is the same for any starting time m . These satisfy

$$P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)}.$$

After n steps you are somewhere, so condition on where and sum. Setting $m = 1$ or $n = 1$ gives the forward and backward equations.

Why $\mathbf{P}^n$ holds the n -step probabilities

Matrix multiplication *is* this sum: the (i, j) entry of $\mathbf{P}\mathbf{P}$ is $\sum_k P_{ik} P_{kj}$, running over every intermediate state k . The routes are mutually exclusive and cover all the cases, so the total is the two-step probability. Repeat, and $\mathbf{P}^n$ covers every path of length n .

With α the distribution of X_0 , the unconditional distribution is $\Pr(X_n = j) = (\alpha \mathbf{P}^n)_j$.

Notation

X_n	the state at time n , random; i, j are particular states
$X(t)$	the general form; in discrete time $X_n = X(t_n)$, $t_n = n$
P_{ij}	one-step probability from i to j , a number
$\mathbf{P}$	the matrix of them; bold marks matrices and vectors
$P_{ij}^{(n)}$	n -step probability, the (i, j) entry of $\mathbf{P}^n$
α	the distribution of X_0 , a row vector

Two places the notation is quietly simplified:

- ▶ The full one-step notation is $P_{ij}^{(n,n+1)}$, carrying the time it applies at. Writing P_{ij} assumes homogeneity.
- ▶ Some notes omit the brackets and write P_{ij}^n for the n -step probability, which literally reads as the number P_{ij} to the power n . Working by hand we will often write it that way too and let context decide. Where a genuine power does turn up it is usually a self-loop, so bracket the base: $(P_{ii})^n$.

Two Steps in Practice

Two coins: coin 1 lands heads with probability 0.7, coin 2 with probability 0.6. Heads today means coin 1 tomorrow, tails means coin 2. Let X_n be the coin used on day n :

$$\mathbf{P} = \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix} \quad \Longrightarrow \quad \mathbf{P}^2 = \begin{pmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{pmatrix}.$$

The top-left entry is Chapman–Kolmogorov in one line: to use coin 1 two days from now, go via coin 1 or via coin 2,

$$P_{11}^{(2)} = P_{11}P_{11} + P_{12}P_{21} = 0.7(0.7) + 0.3(0.6) = 0.67.$$

Unconditional. If day 0 is equally likely to use either coin, $\alpha = (0.5, 0.5)$ and

$$\Pr(X_2 = 1) = 0.5(0.67) + 0.5(0.66) = 0.665.$$

Rows of $\mathbf{P}^2$ still sum to 1, which is the check worth doing.

Exercise 2.1: Matrix warm-up: products and inverses

Exercise 2.1: [dmcpref1, Solution] You are given the following vector and matrices:

$$A = \begin{pmatrix} 3 \\ 1 \\ 10 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 5 & 2 \\ 3 & 12 & 9 \\ 1 & 4 & 7 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 5 & 9 \\ 1 & 3 & 8 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

Calculate the following:

1. $A'B$
2. BC
3. D^{-1}

Simulating a Markov Chain

From state i , the next state is a discrete random variable with probabilities **row i of $\mathbf{P}$** . Apply the Module 1 discrete inverse transform to that row: one uniform, one step.

The two coins, starting in state 1. Row 1 is $(0.7, 0.3)$ and row 2 is $(0.6, 0.4)$, so go to state 1 if $u \leq 0.7$ from state 1, and if $u \leq 0.6$ from state 2:

Step	1	2	3
From state	1	1	2
u	0.43	0.85	0.21
To state	1	2	1

The one new thing: which row you use changes as the chain moves.

When the Process Isn't Markov: Enlarge the State Space

If the next value depends on the last *two* values, X_n on its own is not Markov: knowing the present is not enough.

The fix. Make the state carry enough history. Define

$$Y_n = (X_{n-1}, X_n),$$

and Y_n is Markov. If X_n takes N values then Y_n takes up to N^2 , and the pairs are the states you write $\mathbf{P}$ for.

Two things to check afterwards

Consecutive states must overlap, so from (a, b) you can only reach $(b, \cdot)$: most entries are zero. And the question is usually about X , so translate back at the end, where an event about X_n becomes a set of Y_n states to sum over.

If the dependence reaches back k steps, take k -tuples as states.

Exercise 2.2: Two-day weather chain

Exercise 2.2: [dmc1, Solution] Ross: Example 4.9, Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

If we know that it rained on both Monday and Tuesday, what is the probability that it will rain on Thursday.

Exercise 2.10: Claims process: designing a Markov state space

Exercise 2.10: [dmc9, Solution] An insurance company can receive either 1, 2, or 3 claims per month. The probabilities of the number of claims per month depend on the state of the world. If the last state of the world was:

- “many claims”: the insurer will receive next period either 2 or 3 claims with equal probability;
- “few claims”: the insurer will receive next period either 1 or 2 claims with equal probability.

The current state of world depend on the number of claims:

- If there is 1 claim, the state of the world will be “few claims”,
- if there are 3 claims, the state of the world will be “many claims”;
- if there are 2 claims, the both states of the world occur with equal probability.

1. Define the time index of the Markov process.
2. Define the state space for the Markov process of the aggregate number of claims.

Hint: The state space should be defined such that the process has the Markov property.

Accessibility, Communication, Classes

- ▶ State i is **absorbing** if $P_{ii} = 1$.
- ▶ j is **accessible** from i , written $i \rightarrow j$, if $P_{ij}^{(n)} > 0$ for some $n \geq 0$: any number of steps, not just one. Taking $n = 0$, where $\mathbf{P}^0 = \mathbf{I}$, makes $i \rightarrow i$ automatic.
- ▶ i and j **communicate**, written $i \leftrightarrow j$, if $i \rightarrow j$ and $j \rightarrow i$.
- ▶ The **class** of i is $C(i) = \{j \in S : i \leftrightarrow j\}$.

Communication is an equivalence relation, so the classes partition S : two classes are either identical or disjoint.

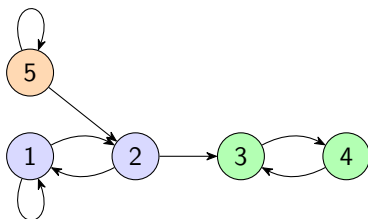
A chain is **irreducible** if there is only one class, so all states communicate.

$P_{ij} = 0$ does not make j inaccessible from i : the route $i \rightarrow k \rightarrow j$ still counts. Follow arrows, don't read off $\mathbf{P}$.

In practice. Draw the arrows, list what each state reaches, group the states that reach each other, then check which classes are closed.

Classes in a Picture

Classification depends only on *which* moves are possible, never on their probabilities. Draw the arrows and ignore the numbers.



Following arrows: 1 and 2 each reach $\{1, 2, 3, 4\}$; 3 and 4 reach only $\{3, 4\}$; 5 reaches everything. The classes are $\{1, 2\}$, $\{3, 4\}$ and $\{5\}$, so the chain is not irreducible. Only $\{3, 4\}$ is **closed**.

Closed or open

A finite closed class is recurrent; an open class, one that can reach states it cannot get back from, is transient. Recurrence comes next week.

Exercise 2.4: Absorbing states, accessibility and classes (parts 1–5)

Exercise 2.4: [dmc3, Solution] For the following transition matrix with states $i = 1, \dots, 7$:

$$\begin{bmatrix} 0 & 0.8 & 0.2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0.6 & 0 & 0 & 0 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

1. Identify the absorbing states;
2. For states $i = 1, \dots, 7$, identify which states are accessible;
3. Identify the communicating classes;
4. Is the Markov chain irreducible?
5. Which states are recurrent and which are transient?

Exercise 2.12: Classify the states of four chains

Exercise 2.12: [dmc11, Solution] Ross: Chapter 4, Q14. For each the following Markov chain:

$$\mathbf{P}_1 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$\mathbf{P}_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{P}_3 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\mathbf{P}_4 = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

1. Specify the classes of the the Markov chains;
2. Determine for all states whether they are transient or recurrent.

Common Mistakes

Not this	This
$P_{ij} = 0$, so $i \not\rightarrow j$	$i \rightarrow k \rightarrow j$ still counts
$\Pr(X_n = j) = P_{ij}^{(n)}$	$\Pr(X_n = j) = (\alpha \mathbf{P}^n)_j$
$\mathbf{P}^n \alpha$	$\alpha \mathbf{P}^n$, row vector on the left
$(P_{ij})^n$	$P_{ij}^{(n)}$, the (i, j) entry of $\mathbf{P}^n$
an answer left in pairs (a, b)	the pairs summed back to X

And every row of $\mathbf{P}$, and of $\mathbf{P}^n$, still sums to 1.

Exercise 2.9: Weather depending on the last three days

Exercise 2.9: [dmc8, Solution] Ross: Chapter 4, Q2. Suppose that whether or not it rains today depends on previous weather conditions through the last three days. Show how this system may be analysed by using a Markov chain. How many states are needed? Define the states.

Thanks for coming!